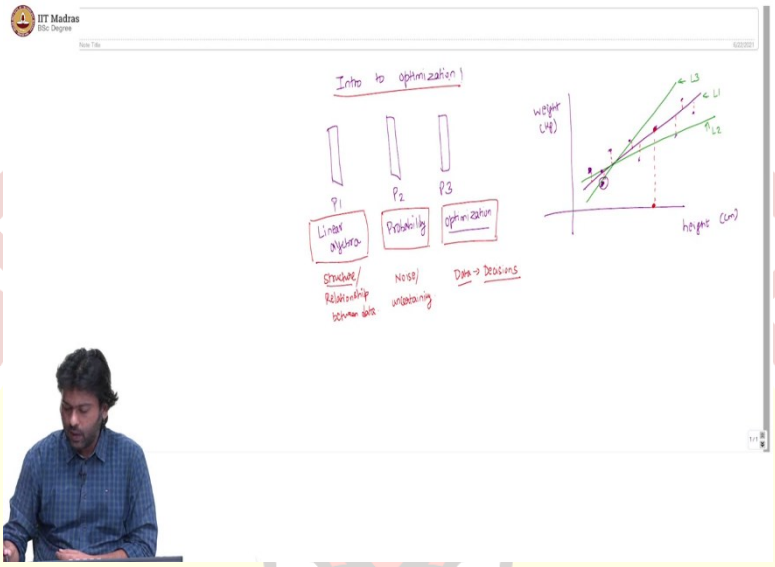


IIT Madras

ONLINE DEGREE

Machine Learning Foundations
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Pillars of machine learning

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Hello, everyone. My name is Arun Rajkumar and I am an instructor for the Foundations of Machine Learning Theory course. Today, we are going to look at introduction to optimization, intro to optimization.

So, what is optimization and why do we care about optimization in a machine learning context? So, according to me there are three pillars of machine learning, rather, so to say, the three foundational pillars of machine learning. I am going to call these pillars P1, P2 and P3 and what are these pillars? So, what are the mathematical pillars on which the theory of machine learning rests?

So, these pillars are linear algebra, probability and optimization. And the focus of my topic would be to look at the introduction to optimization. Now, why do I call these three mathematical areas as the pillars of machine learning for the simple reason that any machine learning problem to understand in detail, you would need some amount of understanding of each of these three pillars. So, let us take a simple example and try to understand the following.

So, let us say you have a data set where, let us say, you are capturing the height and weight of people in a room, height and weight, let us say, height is in centimeters and weight is in kilograms. And now, if we plot this in a two dimensional plane, you are going to get points like

this, so where each point represents one particular person. Now, the question is, if there is a new person who enters this room, and let us say, I know the height of this person, let us say the height was somewhere here. Now, if I want to predict the weight of this person, how would I go about doing this?

Now, of course, we are not going to look at the algorithm or anything of that sort in this course. But let us try to understand what are the fundamental building blocks of this problem. Well, by looking at this graph, one might be tempted to say that there is some relation between the height and the weight. And in particular, this relation seems to be a linear relation. What does that mean? Well, if the height increases, the weight also seem to increase overall on an average and if the height decreases so does the weight and it increases in a proportional way and so we are going to call this a linear relation.

Now, of course, once you have found out what this linear relation is or what is the line that kind of best represents the height-weight relation between people in a room, then given the height of a person, it is going to be very easy to predict the weight. You just find the value that, value corresponding to that height on this particular line. Now, this seems simple, but there are lot of things that have gone in the background to come, to arrive at this particular solution.

The first thing that we assumed here was that the height and weight are linearly related. So, there is a linear relationship between the parameters of interest. And to understand linear relationship between data, one resorts to mathematical foundational ideas in linear algebra. So, this is to understand the structure or relationship between data. So, why linear, why cannot relationship between data be non-linear? Of course, they can be nonlinear as well. And, but the simplest structural relationship between data is a linear relationship.

And it turns out that once you understand the linear relationship very well, in a lot of cases you may also be able to understand the non-linear relationship model and understand the non-linear relationship very well as well. So, the linear algebra part comes because we want to understand the structure in data very well. Of course, again, let us look at the same example. Now, we observe that, well, there is a linear relationship between weight and height agreed, but then it does not really mean that for every point in the data set, which means that for every person whose weight and height we have captured, the weight and height exactly falls on a single line.

So, for instance, if it was the case that all the weight and height points fell on the same line, this point here should actually have been here. None of the points, in fact, in this particular data set fall on the line that I have drawn, yet we believe that well this is a linear relationship. Now,

what does this mean? It means that, overall, there is a linear structure in the data and everything else that is not pertaining to this linear relationship, we are going to think of that as some kind of noise. So, in this particular case, these extra bits are not so significant that they can be considered as noise in the data.

Now, these noise could arise from different sources, because of measurement error, because of incorrect measurements, because of missing measurements, and so on and so forth. So, in some way you not only have to understand the structure in data, but it is also important to understand and model noise or what I will call us uncertainty in data. So, we want to model noise or uncertainty in data. And the mathematical way to deal with uncertainty or noise in data is, of course, the language of probability.

Well, all is good so far. So, we have somehow understood how to deal with structure using the language of linear algebra and how to deal with noise using the language of probability. Now, where does optimization come into picture? So, why should I care about optimization? It seems that, well, if you know the structure, if you can model the uncertainty things are over, well, not quite. To give an example, for the same data set which we think has a linear relationship between height and weight and let us say we have somehow model the uncertainty in data as well, well, there is no single way to describe linear relationship between height and weight.

In other words, there is not a single line. So, there can be infinite possibilities for lines which relate the height and weight. So, this is line one, this is line two, this is line three, and so on. So, you can choose your favorite line and that can maybe, of course, it relates the height and weight. But the question is, is this the line that really relates the height and weight in the data that I actually observed? So, which means that among several possibilities, several possible way of structurally relating the input and output, I care about that way which in some sense best relates the input to the output in this particular example or, in general, best represents the relationship between the data points.

Now, when the moment you have a notion of best, so you want to understand what is the best way to do certain things. Well, that is where you need to come into the world of optimization. So, basically, you have a bunch of lines, infinite possibility of lines, and then among those infinite lines, you are picking one line and saying that, hey, this is the line that really I think represents the relation between height and weight in this data. And how do I decide on this line?

Well, basically, that is where the notion of optimization comes into picture. Basically, optimization is the mathematical tool that allows us to convert data to decisions. Of course, you need structure, you need to model the structure, of course, you need to model the noise or you need to handle uncertainty in one way or other. But more than that, you need to finally convert your data to decisions, and sorry about that, you need to convert data to decisions and the mathematical way to do that is using the language of optimization.

So, with these three pillars, then we can build the palace of machine learning on top of this so to say. So, today's goal and the following few lecture's goal is going to understand the ideas behind optimization from a very basic point of view. We would not really go into the depths of optimization. It is a rich theory in itself with the research done as we speak. But what we are going to understand here is are some kind of basic ideas of optimization which will help us navigate through this world of machine learning and data science. So, with that intro, we will get started.

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